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PHY 101: General Physics I - Mechanics, Matter and Waves

Build the physical intuition and problem-solving habits that General Physics rewards

FACULTY PRIMER

Faculty Primer

Faculty of Physical Sciences
Physics & Industrial Physics | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

This pack treats General Physics as a small number of powerful ideas applied repeatedly. You get a structured method for solving mechanics problems, a clear treatment of simple harmonic motion and waves, and a formula sheet with the conditions under which each formula is valid.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Solve kinematics and dynamics problems using a consistent sign convention and free-body diagrams.
- Apply conservation of energy and momentum to collisions and connected bodies.
- Explain simple harmonic motion and calculate period, frequency and energy.
- Describe wave properties including superposition, interference and standing waves.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Problem-solving protocol

Physics marks are lost in the setup, not in the algebra. Use the same five steps every time until they are automatic.

1. Draw a diagram and label known and unknown quantities with units.
2. Choose a coordinate system and state which direction is positive.
3. Write the relevant principle - Newton's second law, energy conservation, or momentum conservation.
4. Substitute symbols first, then numbers. This makes errors easier to spot and earns method marks.
5. Check magnitude, units and whether the direction of the answer is physically sensible.

> **Free-body diagrams are not optional**
> If two bodies are in contact or connected by a string, draw each body separately. Most 'impossible' pulley problems become simple once the tensions and weights are separated.

Mechanics essentials

Kinematics describes motion; dynamics explains it. Keep them distinct in your answers.

- Equations of motion apply only when acceleration is constant.
- Newton's second law: net force equals mass times acceleration. Net means the vector sum.
- Friction: static friction can take any value up to the limiting value; kinetic friction is roughly constant.
- Work done equals force times displacement in the direction of the force.
- Conservation of mechanical energy applies only when no non-conservative forces do work.
- Momentum is conserved in collisions; kinetic energy is conserved only in elastic collisions.

Oscillations and waves

Simple harmonic motion appears in springs, pendulums, molecules and circuits. Learn the defining relationship and the rest follows.

- Defining condition: acceleration is proportional to displacement and directed toward equilibrium.
- Period of a mass-spring system: T = 2 pi times the square root of m/k.
- Period of a simple pendulum for small angles: T = 2 pi times the square root of L/g.
- Total energy in SHM is proportional to the square of the amplitude.
- Wave speed equals frequency times wavelength.
- Superposition explains interference, beats and standing waves; nodes are points of zero displacement.

> **Angle in radians**
> The small-angle approximation sin(theta) is approximately theta only when theta is in radians. Using degrees silently corrupts the pendulum formula.

Measurement, uncertainty and vectors

Physics begins with measurement and the admission that measurement has limits. Every reading has an uncertainty, and a result cannot be more precise than its inputs. Significant figures are a shorthand for that logic; unit consistency is a check that will catch most errors before they cost marks.

Vectors have magnitude and direction; scalars do not. Displacement, velocity, acceleration and force are vectors; distance, speed, mass and energy are scalars. Resolving a vector into components turns a two-dimensional problem into two one-dimensional problems, which is the single most useful technique in mechanics.